

Leveraging Digital Twins and Generative AI for Effective Urban Mobility Management

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Abstract—Rapid urbanization and population growth have created significant challenges in urban mobility management, including traffic congestion, inefficient public transportation, and environmental pollution. This paper presents the development and implementation of a Digital Twin (DT) for smart mobility, designed to address these issues. The DT platform integrates a diverse range of historical and real-time data, providing a comprehensive view of urban mobility conditions. Descriptive statistics are employed to identify patterns in parking occupancy and violation frequencies, while Machine Learning and Deep Learning algorithms enhance predictive and generative analytics for forecasting parking needs and simulating scenarios with other mobility aspects. Advanced analytics uncover hidden patterns and behaviors, and visualization tools map data onto the urban layout to facilitate spatial planning and resource allocation. Scenario simulation enables urban planners to assess the impact of different strategies in a virtual environment prior to real-world implementation. The integration of Generative Artificial Intelligence (GenAI) models further enhances predictive capabilities and scenario generation. Preliminary results demonstrate the DT platform’s potential in improving urban mobility management, particularly in optimizing parking meter placement and enhancing user experience. Although limited data availability affects long-term prediction accuracy, the model exhibits robustness and adaptability for extended forecasting horizons.

Index Terms—Digital Twin, Smart Cities, Urban Mobility Management, Generative AI, Data-Driven Decision Making

I. INTRODUCTION

The rapid urbanization and growing populations in cities worldwide have led to significant challenges in managing urban mobility, including traffic congestion, inefficient public transportation, and environmental pollution. These issues call for innovative solutions, with smart mobility - a crucial aspect of the smart city concept - offering a way forward by utilizing advanced technologies to build more efficient, sustainable, and user-friendly transportation systems. At the core of these advancements is the Digital Twin (DT), a dynamic, real-time digital replica of physical systems that enables simulation, analysis, and optimization of urban infrastructure. DTs allow city planners and decision-makers to model complex urban systems, predict outcomes under various scenarios, and implement data-driven strategies to enhance city operations.

This paper presents a proof of concept (PoC) for implementing a DT to improve smart mobility. The aim is to develop an

integrated framework encompassing various elements of urban mobility, including parking systems, traffic flow management, and enforcement mechanisms, as shown in Figure 1. The DT is designed to simulate scenarios, analyze results, and provide actionable insights to optimize urban mobility. The PoC was initially tested in the city of Caserta, Campania, Italy, using data from real-time traffic, parking occupancy, public transportation schedules, and environmental sensors. This comprehensive dataset enabled the creation of an accurate DT model, providing valuable insights into Caserta’s urban mobility and laying the groundwork for further experimentation and scenario analysis. The PoC has three main objectives:

- Integration of Urban Mobility Components: Build a holistic framework that models interconnected urban mobility systems.
- Simulation and Scenario Analysis: Use the DT to simulate different urban mobility scenarios, such as peak traffic, special events, and evaluate their impact.
- Data-Driven Decision Making: Derive insights from simulations to support policymakers in making informed decisions.

The paper is organized as follows. Section 2 reviews existing literature on DTs in smart mobility; Section 3 outlines the proposed framework and its components; Section 4 discusses the data sources and processing techniques; Section 5 presents the experimental setup and early results; and Section 6 concludes by summarizing the findings and their implications for smart mobility.

II. RELATED WORK

The application of DT technology in smart cities and smart mobility has garnered substantial attention in recent years.

A. Digital Twins in Smart Cities and Mobility

DTs have emerged as a pivotal technology in the development of smart cities, offering advanced capabilities for real-time monitoring, simulation, and optimization of urban systems. By integrating data from various urban infrastructures, DTs create cohesive and dynamic models of cities, facilitating improved decision-making through scenario analysis and predictive analytics [1]. Recent studies highlight the versatility

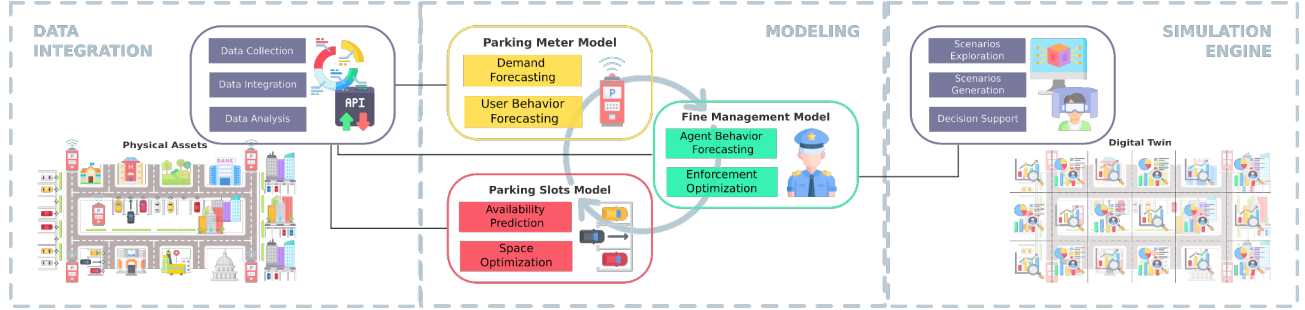


Fig. 1: The proposed Digital Twin (DT) framework for smart mobility management, showcasing three key components: Data Integration, Modeling, and Simulation Engine.

of DTs in urban planning and infrastructure management [2], [3], with applications ranging from improving energy system efficiency and sustainability [4], [5] to enhancing disaster management by simulating emergency scenarios [6], [7]. In the realm of smart mobility, DTs enable detailed simulations and real-time data integration, modeling complex transportation networks for optimized traffic management and reduced congestion [8], [9]. Implementations of DTs in smart transportation systems have demonstrated their potential in optimizing traffic flow and reducing emissions by integrating data from traffic sensors, public transportation schedules, and environmental conditions [10], [11]. Additionally, DTs have been developed for autonomous vehicle networks, improving route planning and reducing energy consumption by leveraging real-time traffic and weather data [12].

B. Digital Twins in Parking and Law Enforcement

Parking management is a crucial component of urban mobility, significantly influencing traffic flow, environmental impact, and overall city efficiency. DTs are increasingly used to optimize parking space utilization, monitor violations, and streamline enforcement processes [13]. One notable application is the development of a DT for an underground parking garage in a smart city [14]. This study combined Wireless Sensor Networks (WSN) with Building Information Modelling (BIM) technologies to enhance environmental management and operational efficiency. This integration enabled dynamic allocation of parking resources based on real-time demand and conditions. Concurrently, law enforcement plays a crucial role in ensuring compliance with traffic and parking regulations while enhancing public safety. DTs support these efforts by providing real-time tracking and analysis of enforcement operations [15]. Through data integration from surveillance cameras, traffic sensors, and social media, DTs improve situational awareness and enable proactive responses to criminal activities [16], [17].

C. Challenges and Gaps

Despite the promising applications of DTs in smart mobility, several challenges and gaps persist [18]. A key challenge is the integration of heterogeneous data sources, which necessitates robust data fusion techniques and the adoption of standardized

data formats [19]. Moreover, the scalability of DT models to cover large urban areas and the computational resources required for real-time simulations remain significant concerns [20]. Additionally, there is a pressing need for more comprehensive studies that incorporate multiple aspects of urban mobility within a unified DT framework. Current research often focuses on individual components in isolation, thereby limiting the potential benefits of an integrated approach.

III. DIGITAL TWIN FRAMEWORK

In this section, we present the Digital Twin (DT) framework aimed at improving smart mobility in Caserta. We provide a detailed overview of the framework's components and objectives, along with the models created for parking meters, parking spaces, agents, and fines.

A. Components

The DT framework consists of interconnected components, each serving a specific purpose:

- (i) Data Integration Layer - Collects and integrates data from various sources such as IoT sensors, traffic cameras, and historical records;
- (ii) Modeling - Employs predictive analytics, ML and DL algorithms to forecast demand and detect patterns;
- (iii) Simulation Engine - Simulates different scenarios to evaluate the impact of various strategies on urban mobility.

Data integration will be addressed in the next section, the primary components of the analytical models are analyzed below.

B. Analytical Models

1) *The Parking Meter Model*: The Parking Meter Model (PMM) is designed to manage and monitor the usage of parking meters throughout the city by integrating real-time data from smart sensors. Its primary functions include:

- Track the status of parking meters - by using data from IoT-enabled parking meters, the model can identify which parking spaces are occupied and which are available at any given moment;
- Analyze historical data and identify usage patterns - this allows for insights into peak usage times, demand forecasting and user behavior.

2) *Parking Slots Model*: The Parking Slots Model (PSM) represents a fundamental component, focusing on optimizing the allocation and utilization of parking spaces throughout the urban landscape. This model integrates data from various sources, including the PMM and additional sensors, to provide insights into: parking availability, facilitate efficient space allocation, and enhance urban mobility.

3) *Fine Management Model*: The Fine Management Model (FMM) uses agent-based modeling to simulate and optimize the enforcement of parking regulations. This approach focuses on the behavior of individual agents, such as drivers and enforcement officers, to understand their interactions and develop effective strategies for managing parking fines. In this model, each enforcement officer is responsible for a designated zone, ensuring that all vehicles have valid parking permits. Officers are required to mark checkpoints within their zones, providing proof of their patrol activities.

4) *Generative Transaction Model*: The Generative Transaction Model (GTM) is an early-stage model designed to simulate parking transaction data by generating synthetic data based on historical trends. Integrated into the simulation engine, the GTM replicates real-world transactions and enables the simulation of various operational scenarios. Its long-term goal is to provide a powerful tool for predictive analysis, offering insights into potential future parking demand and behavior patterns. Initial results from the GTM demonstrate promising alignment with observed transaction data, laying the groundwork for further development.

The models described are key elements of the smart mobility framework, each addressing specific aspects while collaborating seamlessly. The DT acts as a unified ecosystem where these models interact, combining real-time data, predictive analytics, and optimization algorithms. This integration enhances the system's capabilities, providing a comprehensive representation of urban mobility dynamics.

IV. THE DATA

In this section, we present the methodology for data processing, analysis, and simulation, within the DT framework.

A. Processing and Analysis

The core of the DT framework relies on robust data processing and analysis methodologies. This section outlines the techniques and procedures employed in refining, integrating, and analyzing data.

1) *Data Collection*: The first step is the systematic collection of data from various sources. These sources include:

a) *Historical Records*: The study collects data on parking transactions, traffic, and enforcement activities to establish baseline trends and patterns. The primary datasets include the “*transactions history*”, detailing parking meter transactions, and the “*violations history*”, documenting non-payment violations. Supplementary datasets include “*parcometers registry*”, “*parking slots registry*” and “*streets registry*”. The features selected are shown in Table I. After merging these sources, a comprehensive dataframe was created, forming the backbone

Dataset	Feature
transactions history	parcometer id payment type id parking slot number transaction number start park end park
violations history	violation number datetime street id parking slot number house number id vehicle type dispute id officer notes
parcometers registry	parcometer id parcometer coordinates street id
parking slots registry	parking slot number parking slot coordinates street id
streets registry	parking slot number street id street name

TABLE I: Dataset used and data features extracted.

of the DT. All datasets were provided by KCity, the project's collaborating partner.

b) *IoT Sensors*: The above-mentioned data is accessible in real-time through APIs provided by the company, enabling continuous monitoring and analysis. Smart parking meters and traffic sensors deployed throughout the city provide real-time data on parking space occupancy, vehicle counts, and flow.

c) *External Data Sources*: Additional data, such as weather conditions, event schedules, and public transportation usage, are integrated to provide a comprehensive view of factors influencing urban mobility [21].

2) *Data Analysis*: Data analysis is a fundamental aspect of the DT framework, employing various techniques to assess current conditions, predict future trends, and uncover patterns that inform decision-making. The process begins with *descriptive analytics*, which provides a detailed snapshot of urban mobility by summarizing key metrics such as parking occupancy rates, traffic density, and violation frequencies. This helps city planners identify high-demand areas and peak usage periods. Following this, *predictive analytics* utilizes historical and real-time data, supported by machine learning and deep learning models, to forecast future parking demand and traffic conditions. Advanced geospatial visualization tools enhance the analysis by mapping data onto the city's layout, making it easier to identify congestion zones, high-demand parking areas, and enforcement regions for better planning and resource management.

B. Scenarios Simulation

Simulation is a key element of the DT framework, allowing for the exploration of various urban mobility scenarios and

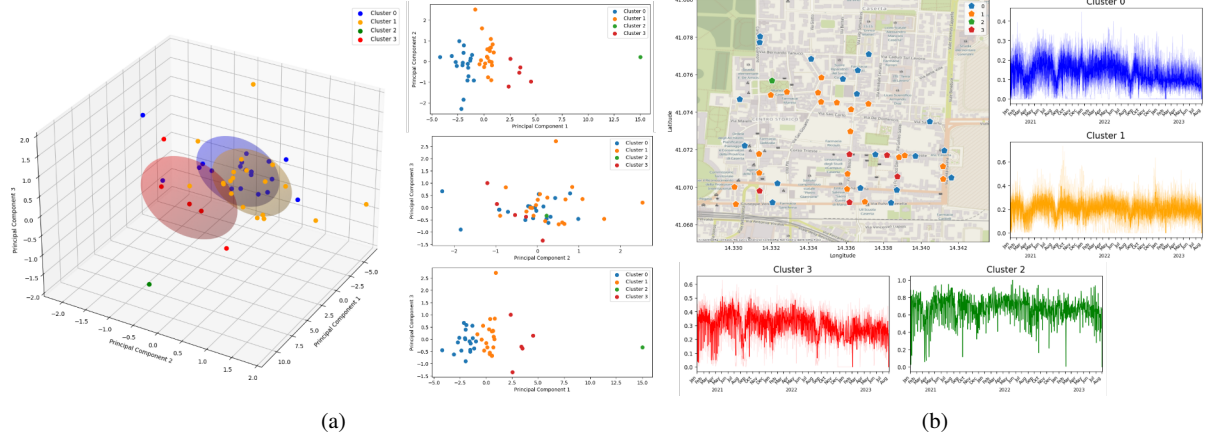


Fig. 2: In (a) results from PCA and TSK-means clustering for four distinct clusters. Ellipsoids delineate the 75th quantile of the density associated with each cluster. In (b) on upper left the map of the Caserta city area displaying the parking meters by cluster. The brighter line represents the Dynamic Time Warping Barycenter Averaging (DBA) within that cluster.

providing insights into the potential impact of different strategies. It helps city planners forecast future conditions and test interventions, such as changes in parking regulations, dynamic pricing, or traffic flow adjustments, in a virtual environment before implementation. By simulating these scenarios, planners can evaluate their effects on mobility and compliance. In *generative analytics*, the integration of GenAI models further strengthens the simulation process by enabling advanced predictive modeling and scenario generation, creating synthetic data for a broader range of scenarios and improving the accuracy and reliability of forecasts.

Overall, the integration of data analysis, simulation, visualization, and GenAI creates a comprehensive decision support system. This system empowers stakeholders to make informed, data-driven decisions, ultimately enhancing urban mobility and improving the quality of life for residents.

V. PRELIMINARY EXPERIMENTAL RESULTS

The implementation of the DT framework for smart mobility in the city of Caserta is a long-term and complex project, involving multiple interconnected models as outlined in section III. This section highlights the early results of the PMM model [III-B1] and the GTM model [III-B4], examining their structures, objectives, and initial findings. These results demonstrate the potential of the DT framework.

A. Descriptive Analytics

The first challenge in our data pre-processing involves the integration of different data sources relevant to predicting urban parking meter usage. We began this process by analyzing the parking meter transaction history.

1) *Spatial influence of Parking Meters*: To optimize system management, we analyzed the distribution of parking meters across the area, calculating the maximum coverage radius for each meter. In Figure 3b, each colored zone represents the coverage area of a parking meter, indicating the distance

within which parking spaces are linked to that meter. This analysis highlights areas for potential improvement, where interventions could enhance coverage and resource efficiency. Additionally, we examined parking usage dynamics to assess system efficiency. Figure 3a presents three visualizations: the 3D graph on the left shows peaks of maximum intensity, indicating the most frequently used areas; the upper right map displays the coverage radius of each meter, revealing zones with higher traffic; and the lower right line graph tracks weekly parking usage, identifying temporal trends and peak periods.

2) *Time Series Clustering*: In this subsection, we examine the time series clustering process using the TimeSeriesKMeans (TSK-means) algorithm [22], an adaptation of the traditional K-means algorithm for time series data. The goal of clustering is to group parking meters with similar transaction patterns while maximizing the differences between clusters. Figure 2a presents the results of Principal Component Analysis (PCA) and TSK-means, while Figure 2b provides a geographical representation of the obtained clusters. This method helps uncover hidden patterns and structures in the data.

3) *Time Series Decomposition*: We use the Seasonal and Trend decomposition method (STL) [23], an effective technique for decomposing time series data into seasonal, trend, and remainder components. STL iteratively estimates the seasonal and trend components while detrending the series through locally weighted scatterplot smoothing (Loess). Applied to parking meter usage data, it reveals key temporal patterns, seasonal variations, and trend differences across areas. This spatio-temporal analysis supports our forecasting models by allowing us to model each component individually and combine them for accurate future usage predictions.

4) *Data Augmentation*: To enhance the robustness and generalisability of our models, we applied data augmentation techniques following data fusion, specifically noise injection, time series decomposition and reconstruction, and time shifting. Gaussian noise was introduced to create small perturba-

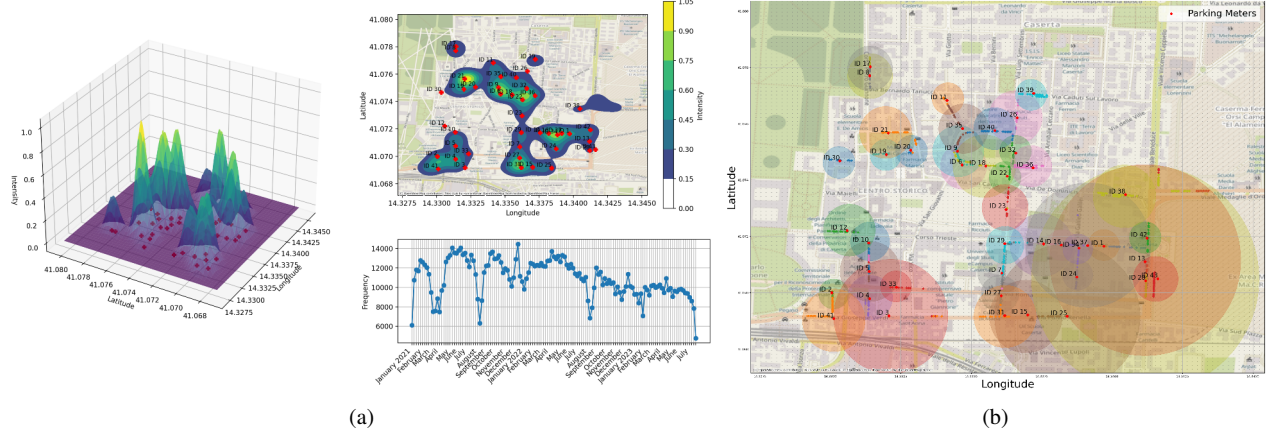


Fig. 3: In (a) dynamics of parking slots use over time. In (b) maximum influence radius per parking meter.

tions, enhancing resilience to input variations. We decomposed the time series data, modified its components slightly, and recombined them to maintain structure while increasing variability. Additionally, time shifting generated new samples by altering the temporal context, further enhancing the model's generalization capabilities.

These preliminary analyses provide an initial in-depth view of the performance of the parking system, supporting the identification of opportunities to further optimize the placement of parking meters and improve the user experience.

B. Predictive Analytics

The Parking Meter Model (PMM) is designed to simulate and analyze parking meter usage across Caserta, utilizing a range of data sources, including historical parking transaction records, real-time sensor data from meters, and external factors like weather and local events. The main goals of the PMM are:

- Forecast parking demand - predict peak usage times and high-demand areas to better allocate resources and manage parking spaces;
- Supporting enforcement strategies - provide information on optimal application areas of parking meters, to maximise coverage and minimise overlaps.

1) *Model Architecture*: A traditional LSTM can forecast n future periods, denoted as $\{\hat{Y}_{t+i}\}_i$, using past time series data $\{Y_{t-j}\}_j$. While this allows the LSTM to learn historical patterns, it limits the incorporation of real-time exogenous variables, such as weather or economic indicators, which can influence predictions. To overcome this limitation, we propose a model that integrates both historical data and exogenous variables, learning a nonlinear function F such that $\{\hat{Y}_{t+i}\}_i = F(\{Y_{t-j}\}_j, \{X_{t+i}\}_i)$. The model consists of two main components: (1) an LSTM module that captures temporal patterns from the historical data, with three LSTM layers applied to components of the STL decomposition, and (2) an Attention Block that processes the exogenous variables.

The LSTM outputs are concatenated and input into an MLP to produce:

$$\{\tilde{Y}_{t+i}\}_i = MLP(Y^{(1)} \oplus Y^{(2)} \oplus Y^{(3)}) \quad (1)$$

where $Y^{(1)}, Y^{(2)}, Y^{(3)}$ are the outputs of the LSTM layers. The Attention Block identifies relevant features from the exogenous variables, yielding energy scores:

$$\{\tilde{X}_{t+i}\}_i = Attention(\{X_{t+i}\}_i) \quad (2)$$

Softmax converts these scores into attention weights, which are applied to compute the context vector for the final prediction.

2) *Experimental Setting*: For our experiments, we employed the Adam optimizer with learning rate of $1e^{-4}$ and weight decay of $1e^{-5}$. The model was trained for up to 6000 epochs, with early stopping implemented after 100 epochs of no improvement. Each batch contained 32 samples, and experiments were conducted on an NVIDIA GeForce RTX 4090 GPU. We selected daily inputs and various forecast horizons, including six exogenous features: a holiday indicator and five weather parameters for daily temperature, precipitation, and wind conditions. We split the data into training and test subsets by forecast horizon. To evaluate the performance, we employed: mean squared error (MSE), mean absolute error (MAE), root mean square error (RMSE), and the coefficient of determination (R^2). Additionally, in line with the real problem, we introduce a custom metric designed for this study, the Absolute Transaction Error (ATE):

$$ATE = \sum_{m=1}^{N_m} (T_m - \bar{T}_m) \quad (3)$$

It represents the difference in parking transactions between the sum of the real values (T_m) and the sum of the model predictions ($\bar{T}_m$).

3) *Main Results and Discussion*: The experimental results confirm the effectiveness of the proposed framework in forecasting across various time horizons. Table II highlights key performance metrics for different time horizons predictions.

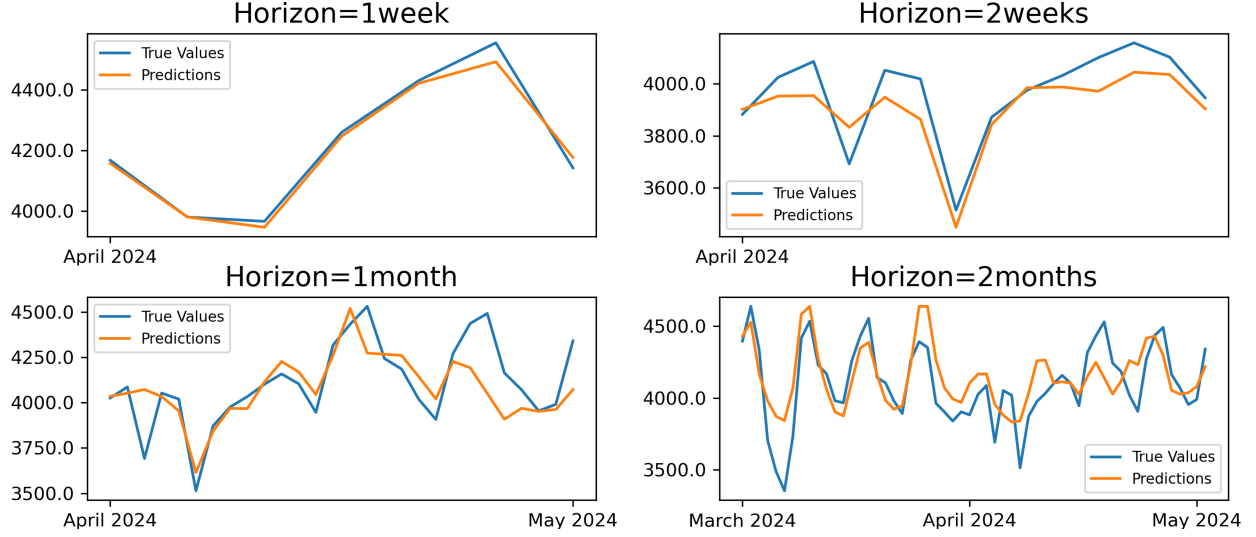


Fig. 4: Predicted and actual parking transactions for different horizons.

Horizon	MSE ↓	MAE ↓	RMSE ↓	R ² ↑	ATE ↓
1 week	0.0003	0.02	0.02	0.97	80
2 weeks	0.006	0.06	0.08	0.69	784
1 month	0.01	0.07	0.10	0.51	758
2 months	0.01	0.09	0.12	0.52	3618

TABLE II: Experimental results across different time horizons.

For a *one-week horizon*, the model shows excellent performance, with low MSE and a high R^2 , alongside minimal ATE. At the *two-week horizon*, the model remains robust, though errors increase for longer forecasts, such as one-month and two-month predictions. Despite higher MSE and RMSE values, the R^2 indicates the model still captures key trends, with ATE values staying within a practical range for urban mobility planning. The limited dataset, spanning from 2021 to August 2023 for training and extending to May 2024 for testing, has impacted accuracy, particularly for longer horizons. Data augmentation has mitigated this to some extent, but predictions will improve as data collection continues. Therefore, these findings are considered preliminary results, reflecting ongoing data collection and refinement efforts to enhance our DT's capabilities. Figure 4 demonstrates a strong alignment between predicted and actual data, underscoring the model's capability to forecast urban mobility trends.

C. Generative Analytics

In this study, we enhanced the COmmon Source COordinated GAN (COSCI-GAN) framework [24] to create synthetic parking meter transaction data under specific conditions. Originally designed to generate multivariate time series, COSCI-GAN captures correlations among individual time series from a common source. Its architecture consists of two primary

components: (1) Channel GANs, where each feature is generated independently using its own generator-discriminator pair, and (2) a Central Discriminator that enforces inter-channel correlations by assessing whether the entire set of generated time series aligns with real multivariate time series data.

1) *Model Architecture*: The COSCI-GAN framework was modified to condition the generation process on specific exogenous factors relevant to parking meter transactions. This allows our model to generate transaction data that reflects real-world influences on parking behavior.

a) *Generator*: Each group-specific generator G_i is an LSTM-based network that takes as input a noise vector $z \in \mathbb{R}^{z_{dim}}$ and a condition vector $c \in \mathbb{R}^{c_{dim}}$, where c_i represents the external conditions (e.g., day, month, temperature, precipitation) for the i -th group. The input $[z, c_i] \in \mathbb{R}^{z_{dim}+c_{dim}}$ passes through an LSTM layer, with the output projected into the target time series dimension via a linear layer. The forward pass is given by:

$$G_i(z, c_i) = \text{Linear}(\text{LSTM}([z, c_i]))$$

b) *Discriminator*: Each group-specific discriminator D_i is also LSTM-based and processes a time series $x_i \in \mathbb{R}^{x_{dim}}$, which can be either real or generated. After passing through an LSTM layer, it outputs a scalar score using a Sigmoid activation, representing the probability that the input sequence is real. The forward pass is:

$$D_i(x_i) = \sigma(\text{Linear}(\text{LSTM}(x_i)))$$

c) *Central Discriminator*: D_{cnt} ensures consistency across the time series generated by different groups. It processes concatenated time series from all groups through an LSTM layer and outputs a decision score via a linear layer. The forward pass is:

$$D_{cnt}(G(z, c)) = \sigma(\text{Linear}(\text{LSTM}(G(z, c))))$$

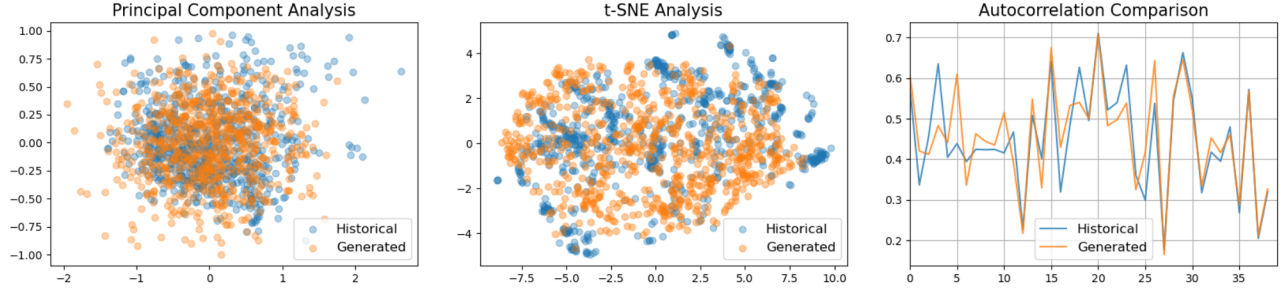


Fig. 5: Comparison of real and generated data using PCA, t-SNE, and ACF, showing alignment in temporal and structural dependencies.

This approach provides a powerful tool for urban mobility planning, enabling the simulation of scenarios and the generation of synthetic data for periods with limited or missing real data. Future enhancements will incorporate additional data, such as violation histories and parking slot registries, to provide a more detailed simulation of parking meter usage.

2) *Experimental Setting*: For our experiments, we used the Adam optimizer with fixed learning rates of $1e^{-4}$ for the generator and central discriminator, and $1e^{-3}$ for the discriminator, with $\beta_{1,2} = [0.5, 0.9]$. The model was trained for up to 1000 epochs with batch size of 64 samples. To evaluate the performance, we analyzed the distributions of real and synthetic data for each parking meter by calculating skewness and kurtosis. Moreover, we examined the autocorrelation of the summed time series across all parking meters. By comparing the autocorrelation between real and synthetic datasets, we can assess whether the model effectively captures the underlying temporal dynamics present in the actual data, validating the quality and reliability of the generated synthetic data.

3) *Main Results and Discussion*: The evaluation of the generated synthetic data was carried out through several comparative analyses with the real parking meter transaction data. Figure 5.1 shows the comparison between the autocorrelation structures of real and generated time series using DBA. The close alignment between the two suggests that the model successfully captures the temporal dependencies across the data, maintaining the sequential patterns present in the historical series. Additionally, the t-SNE analysis in Figure 5.2 demonstrates that the distributions of the generated data are well-aligned with the real data, indicating that the model accurately captures the relationships between the parking meter time series. Finally, Figure 5.3 presents the PCA, where the generated and real data exhibit a strong similarity in their principal components' projections. Although minor deviations are observable, the overall distributional match confirms the model's capacity to replicate key statistical properties of the original dataset. Furthermore, as shown in Figure 6, the PDFs of the historical and generated data for each parcometer confirm the model's ability to recreate realistic distributions. The first graph (linear scale) demonstrates a close alignment between the two datasets, with similar shapes and overlaps.

Skewness (s) and kurtosis (k) values show minimal differences, validating the accuracy of the synthetic data. The second graph (log scale) highlights continued alignment in the tails, indicating the model's ability to capture both common and rare behaviors, which is essential for decision-making in scenarios involving rare events. These results collectively suggest that the generative model not only maintains temporal structure but also effectively simulates the complex interrelations between parking meters, producing realistic synthetic data.

VI. CONCLUSION

The development of a DT framework for smart mobility in Caserta represents a groundbreaking advancement in urban planning and management. By integrating a wide array of data sources, the DT provides a comprehensive and dynamic overview of urban mobility, enabling data-driven decision-making and significantly enhancing the efficiency of urban operations. Our DT framework's application of descriptive and predictive analytics, supported by sophisticated machine learning algorithms, has yielded valuable insights into current and future mobility conditions. The deployment of advanced analytics techniques, such as clustering and anomaly detection, has unveiled hidden patterns and behaviors, deepening our understanding of urban dynamics. In conclusion, the DT framework stands as a powerful tool for enhancing urban mobility and improving the quality of life for residents. The significant results obtained thus far highlight its transformative potential. Continued development, the integration of additional data sources, and advancements in analytical techniques will undoubtedly further strengthen the DT's role in shaping the future of smart cities, paving the way for more efficient, responsive, and sustainable urban environments.

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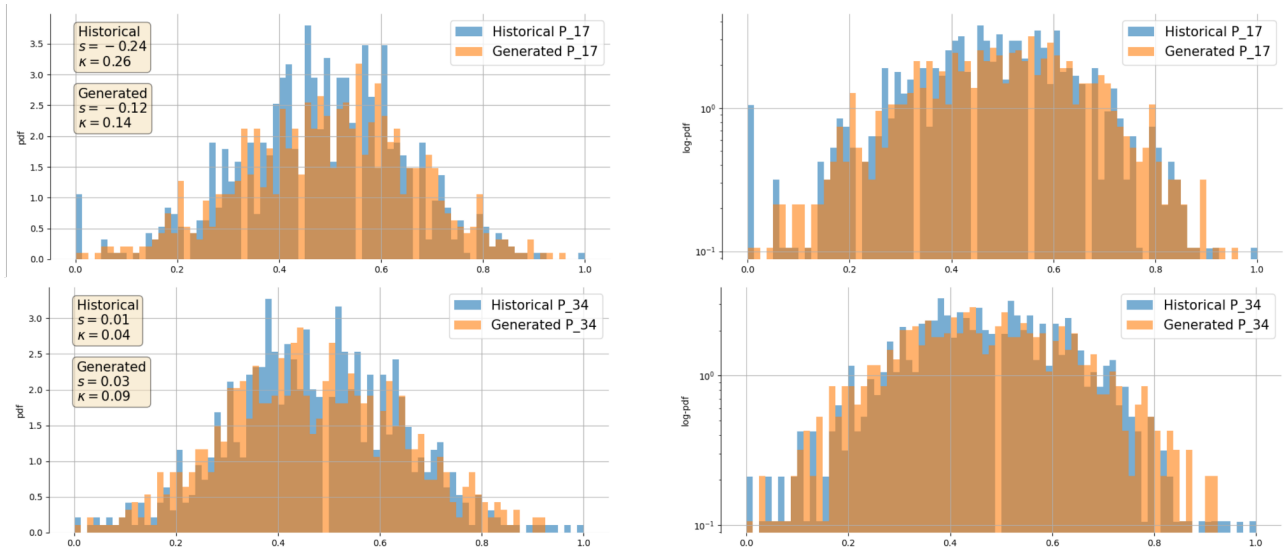


Fig. 6: PDF comparison (linear and log scales) for Parking Meters P17 and P34 as examples, demonstrating the model's ability to replicate both common and rare behaviors.

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